Diagnose ratio misconceptions

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ___________________

1. Read the model: Simplify 45:60 and find an equivalent ratio with first part 12. Copy or complete the first step.

2. A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?

3. Order 0.72, 7/10 and 73%.

4. Miro says: Miro chooses a familiar method before identifying the number type and relationship. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Simplify 45:60 and find an equivalent ratio with first part 12. Copy or complete the first step.
   Answer 3:4 and 12:16.

2. A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?
   Answer 1 3/4 kg.

3. Order 0.72, 7/10 and 73%.
   Answer 7/10, 0.72, 73%.

4. Miro says: Miro chooses a familiar method before identifying the number type and relationship. Is this correct? Explain.
   Answer Name the domain first, represent the relationship, then choose and check the calculation.

Diagnose ratio misconceptions

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ______________

1. Simplify 45:60 and find an equivalent ratio with first part 12.

2. A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?

3. Order 0.72, 7/10 and 73%.

4. Solve this independently and show every stage: Simplify 45:60 and find an equivalent ratio with first part 12.

5. Use a second method or representation for: A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?

6. Check this result using an inverse, estimate or known fact: Order 0.72, 7/10 and 73%.

7. Prove or explain your reasoning: Calculate 5/6 + 7/12.

8. Identify and correct this misconception: Miro chooses a familiar method before identifying the number type and relationship.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Order 0.72, 7/10 and 73%.

Teacher answers and checking prompts

1. Simplify 45:60 and find an equivalent ratio with first part 12.
   Answer 3:4 and 12:16.

2. A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?
   Answer 1 3/4 kg.

3. Order 0.72, 7/10 and 73%.
   Answer 7/10, 0.72, 73%.

4. Solve this independently and show every stage: Simplify 45:60 and find an equivalent ratio with first part 12.
   Answer 3:4 and 12:16.

5. Use a second method or representation for: A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?
   Answer Expected result: 1 3/4 kg. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Order 0.72, 7/10 and 73%.
   Answer Expected result: 7/10, 0.72, 73%. A valid check must be shown.

7. Prove or explain your reasoning: Calculate 5/6 + 7/12.
   Answer 1 5/12. The explanation must show why the method works.

8. Identify and correct this misconception: Miro chooses a familiar method before identifying the number type and relationship.
   Answer Name the domain first, represent the relationship, then choose and check the calculation.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Order 0.72, 7/10 and 73%.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Diagnose ratio misconceptions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: ________________

1. Prove or explain your reasoning: Calculate 5/6 + 7/12.

2. Prove the answer to this question, rather than only calculating it: Simplify 45:60 and find an equivalent ratio with first part 12.

3. Solve this in two different ways and compare them: A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?

4. Work backwards from the answer to reconstruct the question: 7/10, 0.72, 73%.

5. Explain why this reasoning fails: Miro chooses a familiar method before identifying the number type and relationship.

6. Create a new example that tests the same idea as: Order 0.72, 7/10 and 73%.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Calculate 5/6 + 7/12.
   Answer 1 5/12. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Simplify 45:60 and find an equivalent ratio with first part 12.
   Answer Expected result: 3:4 and 12:16. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?
   Answer Expected result: 1 3/4 kg. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 7/10, 0.72, 73%.
   Answer One valid reconstruction is: Order 0.72, 7/10 and 73%. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro chooses a familiar method before identifying the number type and relationship.
   Answer Name the domain first, represent the relationship, then choose and check the calculation.

6. Create a new example that tests the same idea as: Order 0.72, 7/10 and 73%.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.